



Nonlinearity of mechanical damping of a spring-suspended sectional model system and its influence on the vortex-induced vibration response of a bridge deck

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ABSTRACT

This paper investigates the nonlinear characteristics of additional damping provided by various damping adjustment methods or devices and the effects of nonlinear mechanical damping of spring-suspended sectional model (SSSM) systems on the vortex-induced vibration (VIV) responses of bridge decks. First, the characteristics of the additional damping provided by three commonly used damping adjustment methods, such as winding electrical tape on the coil springs, installing eddy current dampers, and tying wire rope rings on the coil springs, to the SSSM systems are studied through free vibration tests. Among these methods, the additional damping provided by tying wire rope rings has the strongest nonlinearity, and the stability of the additional damping provided by winding electrical tape is the worst. Next, based on a wake oscillator model, the effects of the nonlinearity of the structural damping on the lock-in regions and maximum VIV amplitudes of different bridge decks are analyzed. Finally, a method for defining the nominal value of the structural damping ratio is proposed, which effectively mitigates the influence of the nonlinearity of the structural damping on the prediction of the maximum VIV amplitude of a main girder, and its rationality is verified through wind tunnel tests.

Nomenclature

y, A	displacement and amplitude of sectional model
$c_{m,h}, c_{a,h}, c_{n,h}$	original mechanical damping coefficient, windless-air-induced damping coefficient, and total mechanical damping coefficient of SSSM system
$c_{d,h}, \xi_{d,h}$	additional damping coefficient and additional damping ratio of SSSM system provided by damping devices
U, U^*	incoming wind velocity and reduced wind velocity
$k_{n,h}, \omega_{n,h}$	vertical stiffness and natural angular frequency of SSSM system
$y^*(t)$	free decay time history of sectional model
$y_i(t_i), t_i$	continuous positive peaks and corresponding time coordinates in $y^*(t)$, with $i = 1, 2, \dots$
$A^*(t)$	amplitude envelope curve obtained by least squares polynomial fitting of $y_i(t_i)$
$A_i(t_i)$	amplitude on $A^*(t)$ corresponding to t_i
ξ_i	damping ratio corresponding to $A_i(t_i)$
St	Strouhal number
t, τ	time and dimensionless time

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$\xi_{n,h}, \xi_{n,h}(\eta^*), \xi_m$	total mechanical damping ratio, amplitude-dependent total mechanical damping ratio, original mechanical damping ratio, and windless-air-induced damping ratio of SSSM system
$h, \xi_{a,h}$	dimensionless displacement, dimensionless amplitude, and instantaneous dimensionless amplitude of SSSM system
Y, η, η^*	ratio of the Strouhal angular frequency to the natural angular frequency
δ	Strouhal frequency and angular frequency
f_s, ω_s	mass and air-added mass of SSSM system
m, m_a	mass and mass moment of inertia per unit length of SSSM system
m_0, I_0	excitation vibration effect of the shedding vortex
$\tilde{C}_L$	magnitude of $\tilde{C}_L$ on the stationary structure
$\tilde{C}_{L0}$	parameters of a wake oscillator model
H_1^*, G, F	

1. Introduction

Long-span suspension bridges are flexible, lightly-damped structures

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with many closely spaced frequencies. From an aerodynamics perspective, they are bluff bodies, making different bending modes of the main girder susceptible to vortex-induced vibration (VIV) at low wind speeds (Diana et al., 2006; Fujino and Siringoringo, 2013). Large-amplitude VIV can cause long-term fatigue damage to the structure or discomfort to users (Fujino and Yoshida, 2002; Weber and Maślanka, 2012; Gao et al., 2021). Therefore, it is crucial to predict the VIV response of a bridge deck in the stage of wind-resistant design.

Nowadays, both computational fluid dynamics (CFD) and wind tunnel tests are employed to investigate the VIV response of structures. For circular cylinders under low-Reynolds number flows (Martini et al., 2021) or rectangular cylinders with small cross-sectional ratios (Shimada and Ishihara, 2012), CFD, using 2D or 3D simulations, can sometimes successfully predict the onset velocities and responses. However, for bridge decks with complex configurations, predicting VIV responses accurately through numerical simulations remains challenging. Therefore, sectional model tests based on the spring-suspended sectional model (SSSM) system remain the primary method for predicting VIV in long-span bridges. According to similarity principles, the geometry of the bridge deck and the incoming wind conditions should be carefully modeled, which have been demonstrated to have important effects on the vertical VIV responses. Larsen and Wall (2012) studied the influence of shape on the VIV of trapezoidal box girder bridge decks and showed that the side panel/bottom plate angle is a key factor. Laima et al. (2018) conducted experiments to investigate the effects of attachments, such as crash barriers and maintenance rails, on the VIV of a twin-box girder. The effects of the freestream turbulence on the VIV response of typical bluff bodies were also evaluated through wind tunnel tests or CFD (Matsumoto et al., 1993; Kawatani et al., 1999; Daniels et al., 2016). In order to accurately predict the VIV responses of complex girder decks, large-scale sectional models are often utilized in VIV experiments, so that the aerodynamic configuration, especially the details of the attachments can be well captured (Yang et al., 2016). Moreover, high Reynolds numbers can be reached in experiments to reduce the scale effects related to boundary layer thickness (Larsen et al., 2008).

Besides aerodynamic configuration and incoming wind conditions, the dynamic characteristics of a bridge, including modal frequencies and modal shapes, equivalent masses, and modal damping ratios, also influence the VIV responses. Hua et al. (2015) conducted wind tunnel tests to investigate the effect of natural frequencies on the VIV responses of a rectangular cylinder with a cross-sectional aspect ratio of 6 and found that the dimensionless amplitudes and lock-in regions of the vertical VIV remained the same for the suspended system with varying frequencies. Mode shapes influence the distribution of VIV forces along the bridge (Ehsan and Scanlan, 1990), thus resulting in different VIV responses of different modes. This influence is generally considered by employing a mode shape correction coefficient, which relates the VIV response of a sectional model to that of different vertical bending modes of a full bridge. This coefficient is intricately linked with the mode shape function and the mathematical model of vortex-induced forces (VIFs), and has been studied extensively by many researchers (Zhu, 2005; Zhang and Chen, 2011; Zhou et al., 2017; Zhang et al., 2022). It is noted that the oscillation mass and modal frequency of a sectional model are determined from the prototype bridge according to similarity principles, while the modal damping ratio is usually selected based on the wind-resistant design specifications for bridges (JTG/T3360-01-2018, 2018; BS 5400-2, 2006; American Association of State Highway and Transportation Officials, 2017). Therefore, the parameters of the SSSM system should be finely designed to ensure an accurate VIV prediction of the full bridge.

Research on the effect of structural damping and mass on the VIV response of a structure has a long history. According to a review by Marra et al. (2011), for structures such as bridge decks with a mass ratio exceeding 100, the peak VIV amplitude during the lock-in region is determined solely by the mass-damping parameter, specifically the Scruton (Sc) number. The influence of mass-damping parameters on the

VIV responses of cylinders has received widespread attention. For example, Klammo et al. (2006) developed an electromagnetic eddy current damping device to study the effects of damping on the amplitude response of an elastically mounted cylinder. Govardhan and Williamson (2006) conducted a systematic study on the VIV response of a cylinder under varying mass-damping parameters by utilizing a more sophisticated damping control apparatus. This research established a quantitative relationship between the maximum VIV amplitude of the cylinder, the mass damping parameter, and the Reynolds number. On the other hand, studies on the influence of mass-damping parameters on the VIV responses of bridge decks are relatively limited. Much attention has been paid to establishing a relationship between the peak amplitude of vertical VIV with the Sc number for bridge decks with different geometries (Wu and Laima, 2021; Hua et al., 2022), or further to finding an improved mathematical model of VIFs to more accurately predict the variation of the peak amplitude of vertical VIV with the Sc number (Larsen, 1995; Marra et al., 2015).

It is worth noting that there lies an assumption that both the SSSM system and the prototype bridge have the same linear viscous damping in the above studies since a constant damping ratio was always used to define the mass-damping parameters. It has been shown that the mechanical damping of the SSSM system is nonlinear and amplitude-dependent, particularly during large amplitude vibrations (Cao and Ge, 2017). This nonlinearity may have originated from different sources, including material damping, Coulomb friction at joints, and additional damping devices in the SSSM system (Gao and Zhu, 2015). Though nonlinearity exists in the original mechanical damping of the SSSM system, it is usually small. Therefore, damping devices are always needed to provide additional damping for the SSSM system to achieve the 'suggested structural damping ratios' according to the wind-resistant design specifications for bridges. There are four commonly used damping adjustment methods in sectional model wind tunnel tests: installing oil-damped drums, tying wire rope rings on coil springs, winding electrical tape on coil springs, and installing eddy current dampers. Among them, the additional damping provided by the oil-damped drum is very close to linear viscous damping, while the additional damping provided by tying wire rope rings shows obvious amplitude dependence (Gao and Zhu, 2015). Winding electrical tape is simple and convenient, but its damping characteristics have not been deeply investigated yet. Installing eddy current dampers can provide linear viscous damping at low speeds and is easy to adjust quantitatively, but requires special equipment (Huang et al., 2024). Generally, the mechanical damping of a SSSM system can be identified from free decay vibration tests, in which the air-induced nonlinear damping of the oscillating model under zero wind speed should be subtracted carefully (Zhang and Xu, 2019). On the other hand, it has been also demonstrated from the modal identification of some long-span bridges that the damping ratios of certain modes are dependent on the excitation amplitude (Guo et al., 2024; Hwang et al., 2021). For a single-degree-of-freedom structure with nonlinear mechanical damping, the dynamic response will be different from that of a linear one either in the free vibration condition or a forced one (Terenzi, 1999). In the case of post-flutter, the critical flutter speed may also change, due to a nonlinear mechanical damping that increases with the oscillating amplitude (Li et al., 2021). However, to the best of the author's knowledge, the effect of nonlinear mechanical damping on the vertical VIV response of a SSSM is still not clarified.

This paper first explores the nonlinear damping characteristics of large-scale and small-scale SSSM systems for the same open-section deck. Subsequently, the large-scale SSSM system is employed to investigate vertical additional damping provided by three commonly used methods or devices: winding electrical tape on coil springs, installing eddy current dampers, and tying wire rope rings on coil springs. Then, a wake oscillator model is used to study the effect of nonlinear mechanical damping on the amplitude response and lock-in region of vertical VIV. The aerodynamic parameters of this wake oscillator model for the above

open-section deck and a streamlined steel box girder section are identified using the least-squares method. Based on the identification results, the VIV responses of these two main girders with different forms of structural damping are analyzed. Finally, a method for defining the nominal value of the structural damping ratio for predicting the maximum VIV amplitude is proposed, and its reasonableness is verified through sectional model tests.

2. Characteristics of original mechanical damping of SSSM systems

2.1. Free vibration equation for a SSSM system

The SSSM system serves as the primary tool for investigating the VIV responses of bridge decks in long-span bridges. It mainly consists of a rigid sectional model, eight coil springs, a supporting frame, and an additional damping device. In this paper, the mechanical damping of the SSSM system without additional damping devices is referred to as the original mechanical damping. When the damping ratio of the SSSM system is identified using the free decay vibration method, the identification results include not only the original mechanical damping but also the windless-air-induced damping generated by the interaction between the model and the surrounding air in the windless environment. When analyzing the effect of mechanical damping on the VIV response or establishing a mathematical model of VIF, the influence of windless-air-induced damping should be excluded.

Taking vertical vibration as an example, the equation of motion for the free vibration of the SSSM system can be written as:

$$m\ddot{y} + c_{n,h}\dot{y} + k_{n,h}y = -m_a\ddot{y} - c_{a,h}\dot{y} \quad (1)$$

Where m is the mass of the SSSM system, y is the vertical displacement of the model, where the over-dot notation indicates the derivative with respect to time. $c_{n,h} = c_{m,h} + c_{d,h}$ is the total mechanical damping coefficient of the SSSM system, where $c_{m,h}$ is the original mechanical damping coefficient and $c_{d,h}$ is the additional damping coefficient provided by damping devices. $k_{n,h}$ is the vertical stiffness of the SSSM system, m_a is the air-added mass, and $c_{a,h}$ is the windless-air-induced damping coefficient of the SSSM system. According to the research results of [Cao and Ge \(2017\)](#), $c_{m,h}$, $c_{d,h}$, $k_{n,h}$, and m_a , all exhibit a certain degree of amplitude dependency, but the nonlinearity of the mechanical stiffness of the SSSM system is generally weak, and the proportion of air-added mass to the total mass of the model is small. Therefore, this paper focuses on the damping nonlinearity of the SSSM system, without considering the frequency nonlinearity and the influence of air-added mass. As a result, Eq. (1) can be further simplified to:

$$\ddot{y} + 2\omega_{n,h}(\xi_{n,h} + \xi_{a,h})\dot{y} + \omega_{n,h}^2 y = 0 \quad (2)$$

Where $\omega_{n,h} = \sqrt{k_{n,h}/m}$ and $\xi_{n,h} = c_{n,h}/(2m\omega_{n,h}) = \xi_{m,h} + \xi_{d,h}$ are the vertical natural angular frequency and the total mechanical damping ratio of the SSSM system, respectively. $\xi_{m,h} = c_{m,h}/(2m\omega_{n,h})$ is the original mechanical damping ratio, $\xi_{d,h} = c_{d,h}/(2m\omega_{n,h})$ is the additional damping ratio, and $\xi_{a,h} = c_{a,h}/(2m\omega_{n,h})$ is the windless-air-induced damping ratio of the SSSM system.

2.2. Original mechanical damping of SSSM systems

Based on the analysis in Section 2.1, when no additional damping devices are installed, for a bridge deck with a relatively large aspect ratio, the total damping ratio of the SSSM system obtained through the free decay vibration consists of both the original mechanical damping ratio and the windless-air-induced damping ratio. To determine the original mechanical damping ratio of the SSSM system, the model can be rotated 90° for free vibration tests. In this configuration, the aerodynamic performance of the sectional model, characterized by a

substantial aspect ratio, approximates that of a vertical flat plate, effectively reducing the influence of windless-air-induced damping. Thus, the windless-air-induced damping ratio of the SSSM system can be obtained by subtracting the original mechanical damping ratio from the total damping ratio at each amplitude. To obtain the original mechanical damping characteristics of large-scale and small-scale SSSM systems with the same cross-section, a series of steps are taken. Firstly, a free decay vibration test is conducted on the small-scale sectional model in the horizontal state to obtain the total damping of the small-scale SSSM system. Secondly, the small-scale sectional model is rotated by 90° and subjected to a free decay vibration test to obtain its original mechanical damping ratio, allowing for the separation of the windless-air-induced damping ratio. Subsequently, a similar free decay vibration test is performed on the large-scale sectional model in the horizontal state to obtain the total damping of the large-scale SSSM system. Finally, given that the windless-air-induced damping ratio is independent of the scale ratio and mechanical parameters of the model ([Xu and Zhang, 2018](#)), the windless-air-induced damping ratio of the large-scale SSSM system at each dimensionless amplitude is taken from the corresponding value of the small-scale SSSM system. The original mechanical damping ratio of the large-scale SSSM system is then determined by subtracting the windless-air-induced damping ratio from the total damping ratio of the large-scale SSSM system at each dimensionless amplitude. To obtain the variation in damping ratios across different components of the SSSM system with amplitudes, fine damping identification is required. [Xu et al. \(2020\)](#) have thoroughly discussed this process, so this paper will only outline the basic steps of damping identification without delving into the theoretical foundations. The basic steps of damping identification are as follows:

- (1) Use the band-pass filter to eliminate the high-frequency interference signal of the free decay response of the SSSM system and obtain a smoother free decay response $y^*(t)$.
- (2) Extract continuous positive peaks $y_i(t_i)$ on $y^*(t)$.
- (3) Performing a least squares polynomial fitting on the continuous positive peaks $y_i(t_i)$ yields the amplitude envelope curve $A^*(t)$, facilitating the additional reduction of noise interference and signal distortion.
- (4) Calculate the damping ratio ξ_i corresponding to $A_i(t_i)$ on the curve $A^*(t)$ from Eq. (3).

$$\xi_i = \frac{1}{\sqrt{4\pi^2 + \ln^2(A_{i-1}/A_i)}} \ln\left(\frac{A_{i-1}}{A_i}\right) \quad (3)$$

- (5) Based on the amplitude A_i and the corresponding damping ratio ξ_i , obtain the variation of the damping ratio of the SSSM system with its amplitudes.

The bridge deck, modeled using both large-scale and small-scale models, is an open-section deck, as shown in [Fig. 1](#). The basic parameters of the prototype bridge, as well as those of the large-scale and small-scale models, are summarized in [Table 1](#). It can be observed that the bridge deck and models possess a relatively large aspect ratio of 13.9. In conducting free decay vibration tests to identify the original mechanical damping ratio of the SSSM system, it is crucial to eliminate the influence of the windless-air-induced damping ratio. The suspension photographs of both the large-scale and small-scale models within the wind tunnel are illustrated in [Fig. 2](#). The damping ratios of the two systems are identified using the damping identification procedure described above. The trends of their original mechanical damping ratios, windless-air-induced damping ratios, and the sum of both with the dimensionless amplitudes $\eta = A/D$ (A denotes the amplitude.) are presented in [Fig. 3](#).

According to [Fig. 3](#), as the dimensionless amplitudes increase, both the original mechanical and aerodynamic damping ratios of the large-scale and small-scale SSSM systems exhibit a gradual rise, indicating a

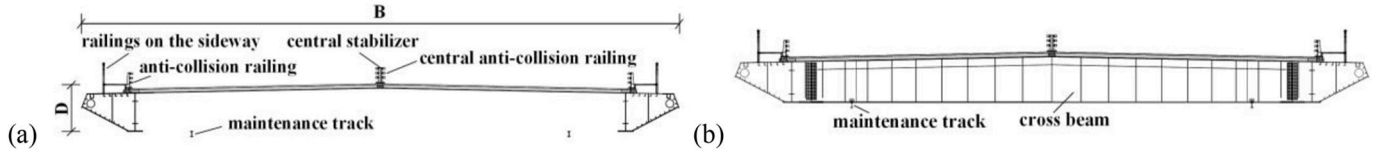


Fig. 1. Cross section of the bridge deck: (a) Without cross beam; (b) With cross beam.

Table 1

Basic parameters of prototype bridge and two sectional models with different scale ratios.

Property	Symbol	Unit	Prototype	Model 1 (1:25)	Model 2 (1:60)
Width	B	m	38	1.520	0.633
Height	D	m	2.735	0.109	0.046
Mass per unit length	m_0	kg/m	40581	64.930	11.273
Mass moment of inertia per unit length	I_0	kg·m ² /m	5759780	14.745	0.444
Vertical frequency	f_h	Hz	0.1008	2.411	1.251
Torsional frequency	f_a	Hz	0.1925	4.517	2.411

strong positive correlation. When the dimensionless amplitude is small, such as $\eta = 0.01$, the original mechanical damping ratios of both the large-scale and small-scale SSSM systems are minimal, approximately 0.03 %. At $\eta = 0.1$, the original mechanical damping ratio of the large-

scale SSSM system increases to around 0.1 %, while the original mechanical damping ratio of the small-scale SSSM system shows no significant change. It is noteworthy that when $\eta < 0.04$, the original mechanical damping ratios of both systems exhibit a trend of fluctuation with amplitudes. Two factors contribute to the fluctuation and inflection points of these original mechanical damping ratios. Firstly, the SSSM system employed in this study has 6 six degrees of freedom. When an artificial excitation is performed along the vertical direction, small disturbances in other directions may also be produced. This disturbance can be ignored at the early stage of free decay vibration when the amplitude of the vertical vibration is large, but may produce noise as the amplitude of the vertical vibration becomes small. Thus, small fluctuation will be induced in the fitting curve of the oscillation envelope, resulting in inflection points in the curve of the original mechanical damping ratio at small amplitudes. Secondly, the energy dissipation mechanism in the connection parts of the model exhibits random behavior under small amplitudes, and the damping components are complex, which may also contribute to this phenomenon.

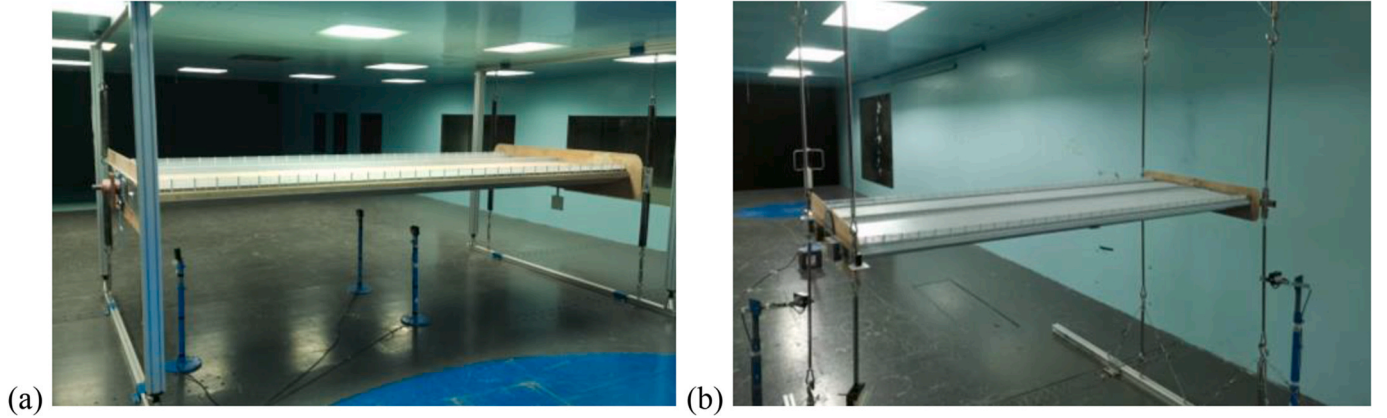


Fig. 2. SSSM systems with two different scale ratios: (a) 1:25 scale ratio; (b) 1:60 scale ratio.

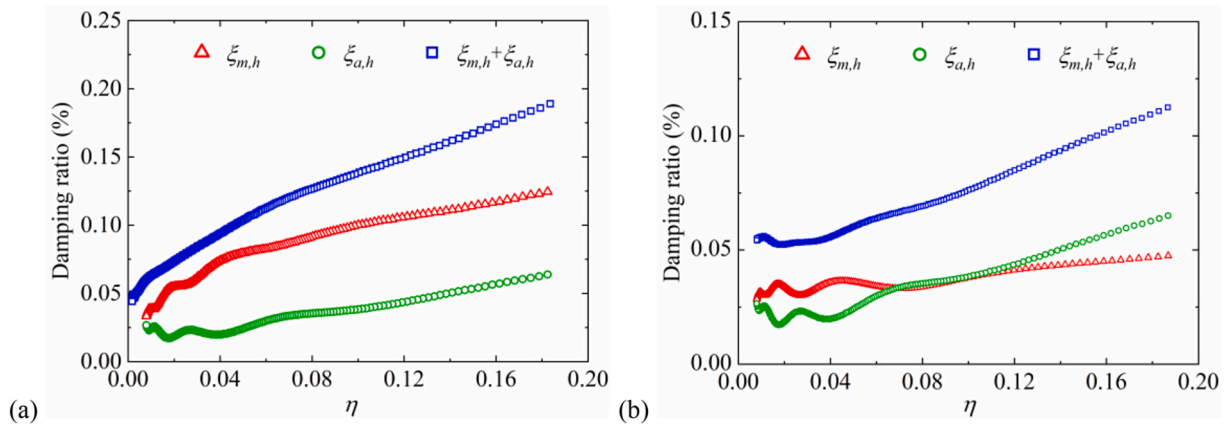


Fig. 3. Damping identification results for SSSM systems with two different scale ratios: (a) 1:25 scale ratio; (b) 1:60 scale ratio.

3. Additional damping characteristics of different damping adjustment methods

This section focuses on the large-scale SSSM system depicted in Fig. 2 (a), primarily comparing the characteristics of the additional damping provided to the SSSM system by three damping adjustment methods: winding electrical tape on coil springs, installing eddy current dampers, and tying wire rope rings on coil springs.

3.1. Additional damping characteristics of winding electrical tape

A photograph of winding electrical tape on the coil springs of the SSSM system is shown in Fig. 4. This section examines the effects of the number of excitations, the number of tape turns, the tape widths, and the tape positions on the additional damping characteristics provided by electrical tape. During winding, the electrical tape is always kept in a tight and undeformed state to avoid pre-pressure exerted on the coil springs. In this way, the additional damping supplied by the electrical tape will be hardly affected by its deformation.

In the static suspension state of the SSSM system, the electrical tape is wound around the center of 8 springs, with a tape width of 5 cm and three tape turns. The variation of the total damping ratios of the SSSM system wound with electrical tape with the number of excitations is presented in Fig. 5. As can be seen, the overall variation trend of the total damping ratios is that it gradually increases with the increase in amplitudes and decreases with the increase in the number of excitations.

The additional damping ratio provided by winding electrical tape is obtained by comparing the total mechanical damping ratios of the SSSM system before and after winding with electrical tape. The variation of the additional damping ratios provided by winding electrical tape with the number of excitations is shown in Fig. 6. As can be seen, the additional damping ratios exhibit an amplitude-dependent characteristic, gradually decreasing with the increase in amplitude. Moreover, with the rise in the number of excitations, the overall trends of the additional damping ratios are decreasing, while their amplitude-dependent features remain fundamentally stable. For example, for the first excitation (test 1), at dimensionless amplitudes of 0.02 and 0.10, the additional damping ratios are 0.27 % and 0.24 %, respectively, a decrease of 0.03 %. After ten excitations, with a dimensionless amplitude of 0.10, the additional damping ratio drops to 0.20 %, representing a reduction of 0.04 %. The decrease in the additional damping ratios provided by winding electrical tape as the number of excitations increases can be



Fig. 4. Damping adjustment method of winding electrical tape.

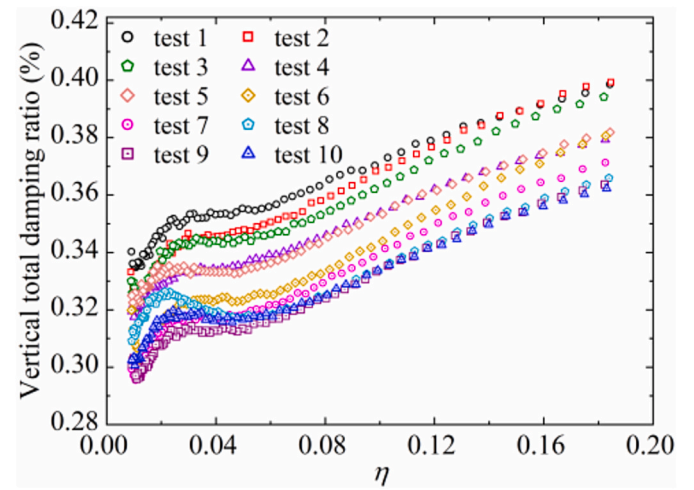


Fig. 5. Total damping ratios of the SSSM system wound with electrical tape versus the number of excitations.

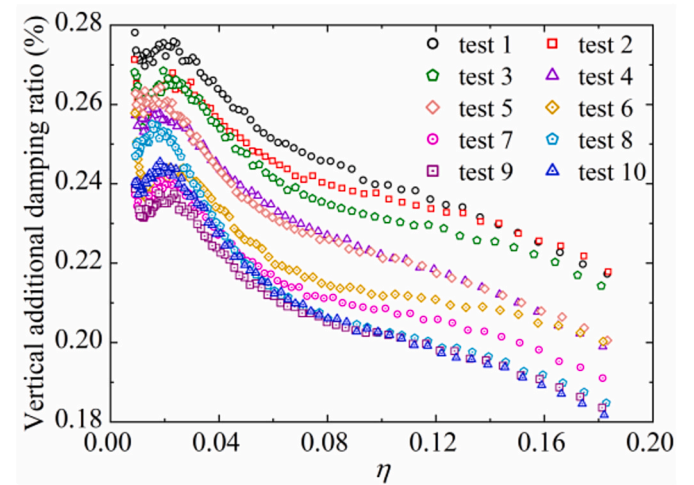


Fig. 6. Additional damping ratios of winding electrical tape versus the number of excitations.

explained from two aspects: the viscosity and deformation of the tape. With regard to viscosity, increased excitations weaken the adhesive force. Vibrations cause molecular rearrangements, which diminish bonding strength, reducing adhesion and energy transfer. This leads to a decrease in additional damping ratios. In terms of deformation, repeated stretching and compression from multiple excitations induce fatigue. This fatigue impairs the tape's ability to recover its original shape, resulting in a diminished capacity to store and dissipate energy. As shown in Fig. 6, as the number of excitations increases, the additional damping ratios gradually turn to be stable. Therefore, several rounds of excitations can be performed in advance to improve the stability of the additional damping ratios during wind tunnel tests.

The variation of the additional damping ratios provided by winding electrical tape with the number of tape turns is shown in Fig. 7. The electrical tape, wound around the center of each spring with a 5 cm width, is applied in 1, 2, and 3 turns. Each condition is excited five times, and the average result is taken as the representative value of the additional damping ratios under a certain amplitude. As can be seen, with an increase in the number of tape turns, the additional damping ratios gradually rise, but the rates of increases gradually diminish. The amplitude-dependent curves of the additional damping ratios at different numbers of tape turns are largely parallel. This suggests that

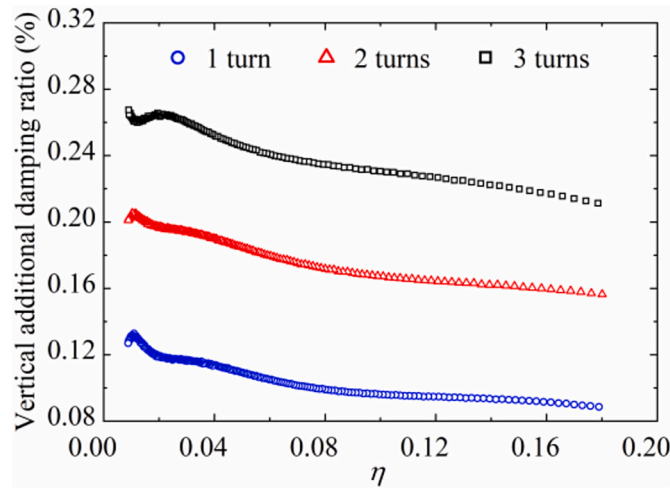


Fig. 7. Additional damping ratios of winding electrical tape versus the number of tape turns.

additional turns primarily enhance the linear viscous damping of the electrical tape, while it barely increases the amplitude-dependent characteristics of the additional damping ratio. For example, at a dimensionless amplitude of 0.10, the additional damping ratios for 1, 2, and 3 tape turns are 0.10 %, 0.17 %, and 0.23 % respectively. The incremental increases in additional damping ratios for the 2nd and 3rd turns are 0.07 % and 0.06 % respectively.

As shown in Fig. 8, the electrical tape is wound around the center of each spring with one turn, and the tape widths used are 5 cm, 9 cm, and 13 cm, respectively. This is done to investigate the effects of tape widths on the additional damping ratios. The variation of the additional damping ratios provided by winding electrical tape with tape widths is shown in Fig. 9. As can be seen, the additional damping ratios gradually increase with tape widths, and the increments of the additional damping ratios are roughly proportional to the increases in tape widths. The amplitude-dependent curves of the additional damping ratios for different tape widths are basically parallel. This indicates that increasing the tape width mainly enhances the linear viscous damping provided by electrical tape, while it hardly increases the amplitude-dependent

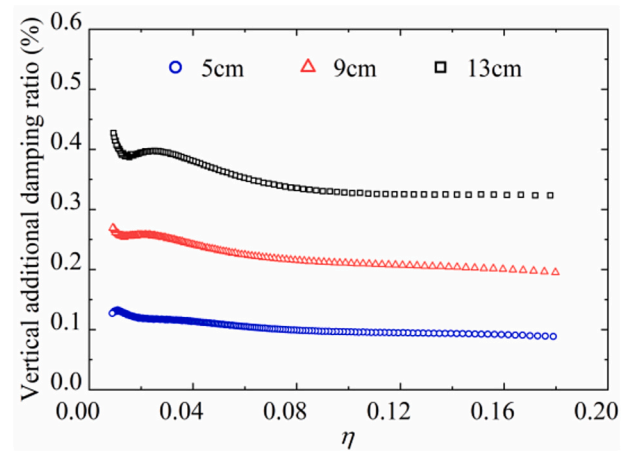


Fig. 9. Additional damping ratios of winding electrical tape versus tape widths.

characteristics of the additional damping ratio. For example, at a dimensionless amplitude of 0.10, the additional damping ratios for tape widths of 5 cm, 9 cm, and 13 cm are 0.10 %, 0.21 %, and 0.33 %, respectively. The increases in damping ratio per additional 4 cm of tape width are 0.11 % and 0.12 %, which are nearly the same.

As shown in Fig. 10, the electrical tape is wound at the top, middle, and bottom of each spring, respectively, with one turn of tape and a width of 5 cm. This is done to investigate the effects of tape positions on the additional damping ratios. The variation of the additional damping ratios provided by winding electrical tape with tape positions is shown in Fig. 11. As can be seen, the maximum additional damping ratio is achieved when the tape is wound at the top, while the minimum occurs when it is wound in the middle. Apart from the fluctuation at small amplitudes, the amplitude-dependent curves of the additional damping ratios for different tape positions are basically parallel. This indicates that changing tape positions mainly causes the change of the linear viscous damping provided by the electrical tape. For example, at a dimensionless amplitude of 0.10, the additional damping ratios provided by the electrical tape for the top, middle, and bottom positions are 0.13 %, 0.10 %, and 0.11 %, respectively.

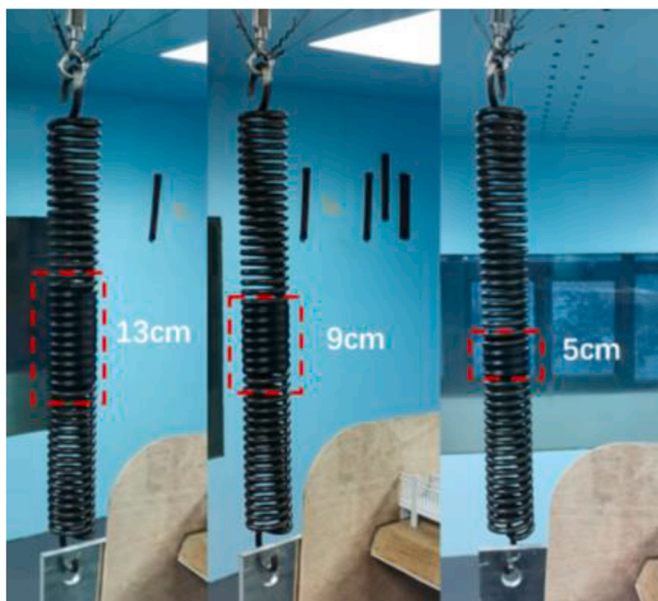


Fig. 8. Damping adjustment method of winding electrical tape for different tape widths.

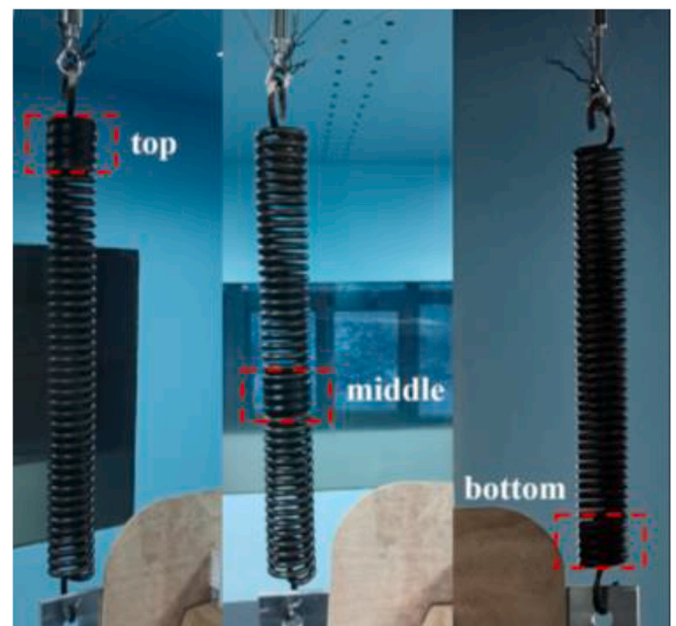


Fig. 10. Damping adjustment method of winding electrical tape for different tape positions.

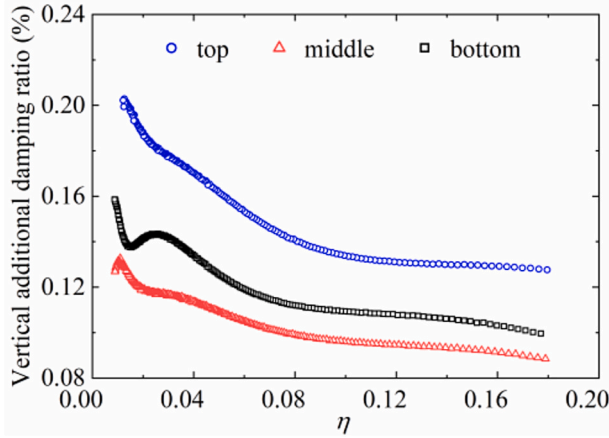


Fig. 11. Additional damping ratios of winding electrical tape versus tape positions.

In summary, winding electrical tape around the coil springs effectively enhances the additional damping ratio of the SSSM system. Specifically, this value is mainly influenced by the number of tape turns, tape widths, and tape positions, and it gradually decreases with an increase in the number of excitations. It is worth noting that the additional damping provided by the electrical tape primarily stems from the linear viscous damping generated by its viscoelastic deformation. Furthermore, it encompasses other complex damping mechanisms such as friction, leading to a gradual decrease in the equivalent damping ratio as the amplitude increases, exhibiting an amplitude-dependent characteristic. On the other hand, increasing tape widths and the number of tape turns is advantageous for elevating the proportion of additional viscous damping, consequently reducing the nonlinear characteristics of the additional damping.

3.2. Additional damping characteristics of installing eddy current dampers

The eddy current damper consists of 8 permanent magnets, a conductor plate, two back irons, an air-gap adjustment device, and a base, as shown in Fig. 12(a). The conductor plate is fixed to the cross-beam of the sectional model and vibrates with it. The permanent magnets and backing irons are symmetrically arranged on both sides of the conductor plate and connected to the air-gap adjustment device, which changes the distance between them and the conductor plate to adjust the damping coefficient. The air-gap adjustment device is fixed to the base

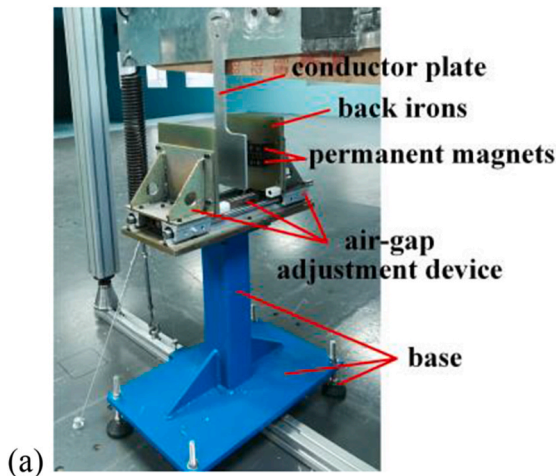


Fig. 12. Installation photo of eddy current dampers: (a) basic structure of the device; (b) installation on the SSSM system.

attached to the wind tunnel wall. The dimensions of the conductor plate, permanent magnet, and back iron are 140 mm × 130 mm × 6 mm, 40 mm × 25 mm × 20 mm, and 170 mm × 190 mm × 10 mm, respectively. The air gap between the permanent magnets and the conductor plate can be manually adjusted from 5 mm to 50 mm to control the additional damping. More details about the eddy current damper can be found in Huang et al. (2024). The installation photo of eddy current dampers is shown in Fig. 12(b). Two eddy current dampers are installed diagonally at the left and right ends of the model to provide additional damping ratios for both the vertical and torsional modes of the SSSM system. During the sectional model tests, the damping coefficients of the two vertical dampers are adjusted to be the same and installed in a centrally symmetrical arrangement. Therefore, during vertical vibration, the damping force of the two eddy current dampers is equal in both magnitude and direction, which results in a purely vertical damping force. However, it should be noted that there is an additional force in the up-and-down mode of the sectional model for the torsional VIV, which is beyond the scope of this paper. In this section, based on the vertical bending mode, the influence of the number of excitations and the air gaps on the additional damping characteristics of installing eddy current dampers will be studied.

To investigate the influence of the number of excitations on the additional damping of eddy current dampers, the air gap between the permanent magnets and the conductor plate was set to 22 mm, and 10 free-decay vibration tests were conducted. The variation curve of the

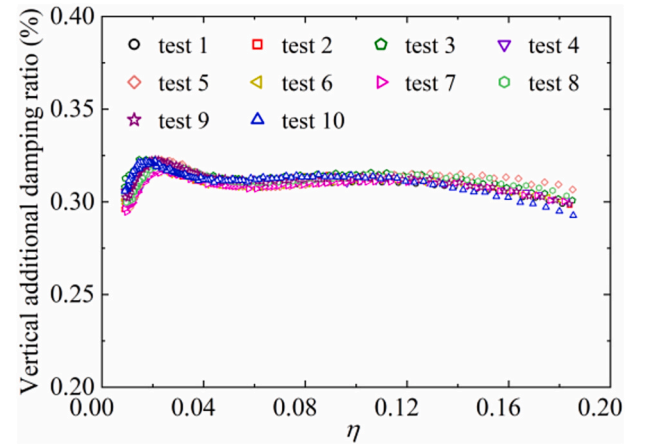


Fig. 13. Additional damping ratios of installing eddy current dampers versus amplitudes under different excitation times.

additional damping ratio with the amplitude for each excitation is shown in Fig. 13. As can be seen, the curve of the additional damping ratio with the amplitude for each excitation basically overlaps. Notably, the additional damping ratio remains essentially constant with the amplitude, with only slight fluctuations at the end. This indicates that the eddy current dampers provide stable and nearly linear viscous damping for the SSSM system.

By reducing the air gap between the permanent magnets and the conductor plate, the damping coefficient of eddy current dampers can be increased, providing a larger damping ratio for the SSSM system. Five air gaps of 13.5 mm, 14.5 mm, 18.3 mm, 22.0 mm, and 29.5 mm were selected for the test, and 10 free decay vibration tests were conducted for each air gap. The average value of 10 identification results is taken as the representative value of the additional damping ratio under a certain amplitude, and the results are shown in Fig. 14. As can be seen, the additional damping ratios provided by installing eddy current dampers do not vary with amplitudes under different air gaps. As the air gaps decrease, the additional damping ratios increase, reaching approximately 1.1 % at an air gap of 13.5 mm. This demonstrates that the eddy current dampers enable fine, continuous, and wide-range adjustment of the total mechanical damping ratios for the SSSM system.

3.3. Additional damping characteristics of tying wire rope rings

The primary energy dissipation mechanism of tying wire rope rings is friction between the wire rope strands. To investigate the additional damping characteristics of tying wire rope rings, each of the four springs in the upper part of the SSSM system was tied with a wire rope ring, 400 mm in diameter, as shown in Fig. 15. Each rope ring consists of two wraps of wire rope with a 4 mm diameter. Five free decay vibration tests were conducted, and the curves of the additional damping ratios with amplitudes corresponding to each test are shown in Fig. 16. As can be seen, the curves of the additional damping ratios with amplitudes for different tests are almost completely coincident, indicating that the additional damping provided by tying wire rope rings is stable and unaffected by the number of excitations. When the dimensionless amplitude $\eta > 0.02$, the additional damping ratio decreases rapidly as the amplitude increases. For example, at dimensionless amplitudes $\eta = 0.02$ and $\eta = 0.16$, the corresponding additional damping ratios are 0.45 % and 0.20 %, respectively. This reduction of over 50 % demonstrates that the additional damping ratio exhibits a significant amplitude-dependent behavior. It should be noted that although the wire rope rings are constantly deforming during the vibration of the model, which has the effect of adding stiffness, its impact on the frequency of the SSSM system is still very small (Gao and Zhu, 2015). In addition, the main damping provided by winding electrical tape and installing eddy current dampers

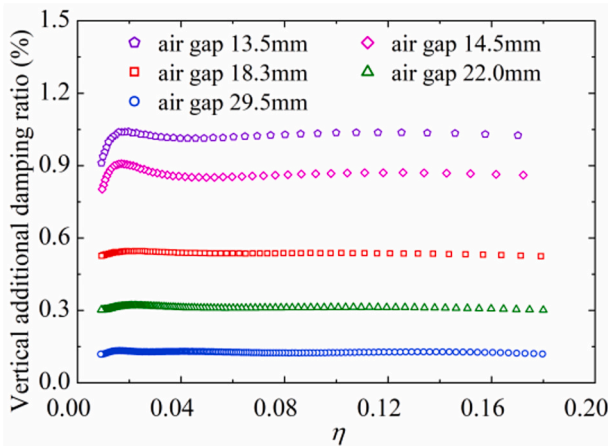


Fig. 14. Additional damping ratios of installing eddy current dampers versus amplitudes under different air gaps.

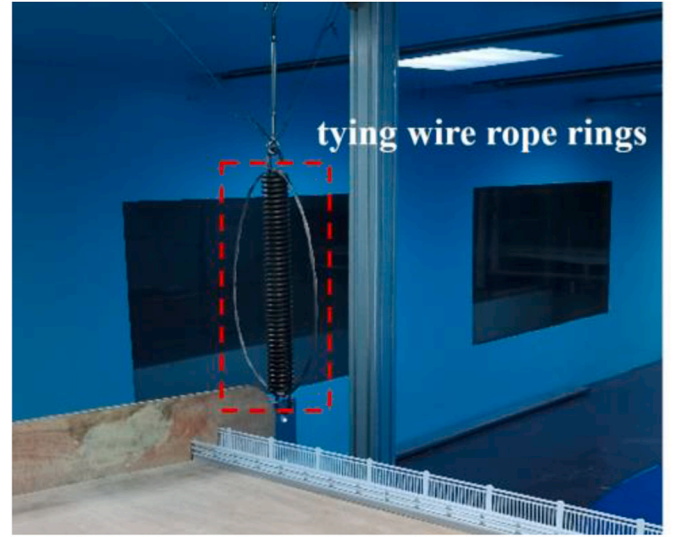


Fig. 15. Damping adjustment method of tying wire rope rings.

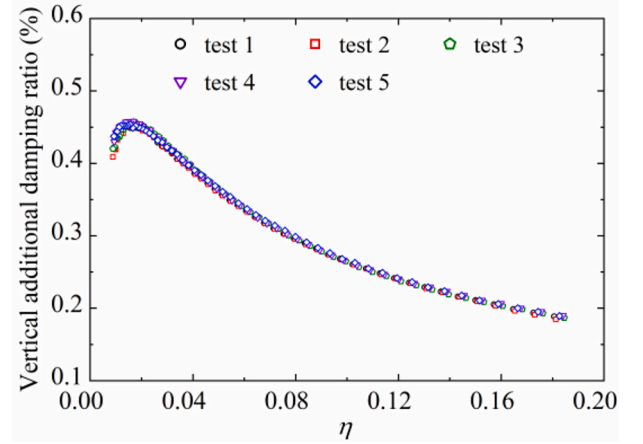


Fig. 16. Additional damping ratios of tying wire rope rings versus amplitudes under different excitation times.

is still viscous damping, and its influence on the frequency of the SSSM system can be ignored.

4. Effects of the nonlinearity of structural damping on the VIV of main girders

Compared to single-degree-of-freedom mathematical models of VIVs, the wake oscillator model can simulate the lock-in region of the main girder. Therefore, this section employs the wake oscillator model to study the effects of the nonlinearity of the structural damping on the vertical VIV responses of main girders, as well as to determine a reasonable representative value for the amplitude-dependent structural damping ratio. Considering that no measured data on the amplitude-dependent structural damping ratios of the prototype bridges is available, the analysis is conducted with reference to the nonlinear characteristics of the total mechanical damping in the SSSM system. It's important to note that this same approach can be extended to other damping models.

4.1. Wake oscillator model

There are various types of wake oscillator models. This paper adopts the wake oscillator model proposed by Facchinetti et al. (2004), whose

applicability for modeling the VIV response of main girders was analyzed by Xu et al. (2015). The model employs the following dimensionless expressions:

$$\begin{cases} Y'' + 2\xi_{n,h}(\eta^*)Y' + Y = M\delta^2\tilde{C}_L + 2\pi M\delta St H_1^* Y' \\ \tilde{C}_L'' - \delta G\tilde{C}_{L0}^2\tilde{C}_L + 4\delta G\tilde{C}_L^2\tilde{C}_L + \delta^2\tilde{C}_L = f(Y, Y', Y'') \end{cases} \quad (4)$$

Where $Y = y/D$ is the dimensionless displacement, $(\cdot)$ means deriving it with dimensionless time $\tau = \omega_{n,h}t$. $\xi_{n,h}(\eta^*)$ is the amplitude-dependent total mechanical damping ratio, which is a function of the instantaneous dimensionless amplitude $\eta^* = \sqrt{Y^2 + (Y'/\delta)^2}$. $\delta = \omega_s/\omega_{n,h}$ denotes the ratio of the Strouhal angular frequency to the natural angular frequency, which can also be expressed as $\delta = St \cdot U^*$, $St = (f_s D)/U$ is the Strouhal number, U is the incoming wind velocity, $U^* = U/(f_{n,h}D)$ is the reduced wind velocity. $M = \rho D^2/(8\pi^2 m S^2)$ is the mass parameter, where ρ is the air density. $\tilde{C}_L$ is the excitation vibration effect of the shedding vortex, and $\tilde{C}_{L0}$ is the magnitude of $\tilde{C}_L$ on the stationary structure. $f(Y, Y', Y'')$ represents the effect of model vibration on the wake flow and can be taken as FY , FY' , or FY'' , which denote displacement coupling, velocity coupling, or acceleration coupling, respectively. H_1^* , G , and F are model parameters.

4.2. Analysis conditions for nonlinear structural damping

For SSSM systems, structural damping refers to total mechanical damping. In the wind-resistant design specification for highway bridges of China (JTG/T3360-01-2018, 2018), the calculation formula for the dimensionless value h_0 , which represents the amplitude limitation of VIV for each vertical bending mode of the main girder, is given as $h_0 = 0.04/(f_h D)$. In this paper, the amplitude-dependent structural damping ratio at h_0 is defined as the 'specification damping ratios' $\xi_{n,h0}$, which represents a constant structural damping ratio that can be used in sectional model tests when the amplitude-dependent characteristic of the total mechanical damping ratio is ignored. In this paper, for the prototype bridge, the first-order vertical bending frequency is $f_h = 0.1008\text{Hz}$, corresponding to a dimensionless value $h_0 = 0.04/(f_h D) = 0.145$ for vertical VIV. In addition, the amplitude-dependent structural damping ratio at $h_0/4$ is also selected as another possible way to linearize the total mechanical damping ratio of the SSSM system, which is hereinafter referred to as the 'small-amplitude damping ratio' $\xi_{n,1/4h0}$. Five representative total mechanical damping ratios are selected for analysis from three damping adjustment methods, including winding electrical tape, installing eddy current dampers, and tying wire rope rings. For the first two methods, two damping conditions are selected each, with their 'specification damping ratios' $\xi_{n,h0}$ approximately 0.3 % and 1.0 %, respectively. For the wire rope rings method, one damping condition is selected with a 'specification damping ratio' $\xi_{n,h0}$ close to 0.3 %. Table 2 presents the 'specification damping ratios' $\xi_{n,h0}$ and 'small-amplitude damping ratios' $\xi_{n,1/4h0}$ for these damping conditions. The amplitude-dependent curve for each damping condition is shown in Fig. 17. It can be observed that damping conditions 1 and 2 exhibit relatively weak nonlinearity, with condition 1 approximating a constant damping ratio. This occurs because the original mechanical damping ratio of the SSSM system increases with amplitudes, in contrast to the

amplitude-dependent behavior of the additional damping provided by winding electrical tape. Damping conditions 3 and 4, produced by installing eddy current dampers, show moderate damping nonlinearity, while condition 5, from tying wire rope rings, demonstrates the highest level of damping nonlinearity.

4.3. Effects of the nonlinearity of structural damping on the VIV of a Π -shaped composite girder

The vertical VIV responses of the Π -shaped composite girder under two levels of structural damping are shown in Fig. 18. It is important to note that the results are based on experimental data from a sectional model and do not account for the mode shape correction coefficient of the prototype bridge. From the characteristics of the curves in Fig. 18, it is recommended to adopt the acceleration coupling type in Eq. (4), i.e., $f(Y, Y', Y'') = FY''$ (Facchinetti et al., 2004). By combining the original mechanical damping ratio $\xi_{m,h}$ in Fig. 3(a) and key structural parameters $St = 0.087$, $M = 0.0003749$, and $\tilde{C}_{L0} = -0.69$ in Eq. (4), the experimental data is fitted using the least-squares method, yielding $H_1^* = -42.9987$, $G = 0.0395$, and $F = 13.8001$. The comparison between the fitted curves and the experimental data is shown in Fig. 18. As can be seen, although the fitted curves do not completely coincide with the experimental data, they effectively simulate the maximum VIV amplitudes and lock-in regions of the Π -shaped composite girder under different levels of structural damping.

Combining Eq. (4) and the parameters identified above, the vertical VIV responses of the Π -shaped composite girder under each damping condition in Fig. 17 are calculated using the 4th-order Runge-Kutta method. The calculation results are displayed in Fig. 19. As can be seen, under all damping conditions, the lock-in regions corresponding to both amplitude-dependent structural damping and constant structural damping are nearly identical. This indicates that the damping nonlinearity has little effect on the lock-in region of the Π -shaped composite girder studied in this paper. In Fig. 19(a), (c), and 19(e), it is notable that within the lock-in regions, dramatic increases and decreases in the VIV amplitude occur near both the initial and cut-off wind speeds. This phenomenon occurs due to the non-monotonic variation of the aerodynamic damping ratios with amplitudes under these specific wind speeds and the significant difference in the variation of aerodynamic damping ratios with amplitudes under adjacent wind speeds (Zhang et al., 2019; Wang and Chen, 2022).

For the maximum VIV amplitudes, the results corresponding to the amplitude-dependent structural damping and the constant structural damping under damping condition 1 are basically the same, due to the weak damping nonlinearity in this condition. However, under the damping conditions 2 through 5, notable differences emerge between the maximum VIV amplitudes for the amplitude-dependent and the constant structural damping. In particular, under the damping condition 5 with the strongest damping nonlinearity, the maximum VIV amplitudes corresponding to the amplitude-dependent structural damping ratio $\xi_{n,h}$, the 'specification damping ratio' $\xi_{n,h0}$, and the 'small-amplitude damping ratio' $\xi_{n,1/4h0}$ are 0.068, 0.080, and 0.059 respectively. The calculation errors for the corresponding constant structural damping ratios, i.e., $\xi_{n,h0}$ and $\xi_{n,1/4h0}$, are 17.6 % and 13.2 % respectively. Therefore, when predicting the VIV responses of the main girder, it is crucial to account for significant nonlinearity in the structural damping of the prototype bridge.

4.4. Effects of the nonlinearity of structural damping on the VIV of a streamlined box girder

The effects of the nonlinearity of structural damping on the VIV responses of main girders may vary depending on section types. Combining the wake oscillator model of a streamlined box girder fitted by Xu et al. (2015) and the amplitude-dependent structural damping

Table 2

Damping parameters of each damping condition under different damping adjustment methods.

Damping condition	Damping adjustment methods	$\xi_{n,h0}$	$\xi_{n,1/4h0}$
Damping condition 1	Winding electrical tape	0.34 %	0.33 %
Damping condition 2	Winding electrical tape	0.97 %	1.04 %
Damping condition 3	Installing eddy current dampers	0.28 %	0.24 %
Damping condition 4	Installing eddy current dampers	0.99 %	0.96 %
Damping condition 5	Tying wire rope loops	0.31 %	0.46 %

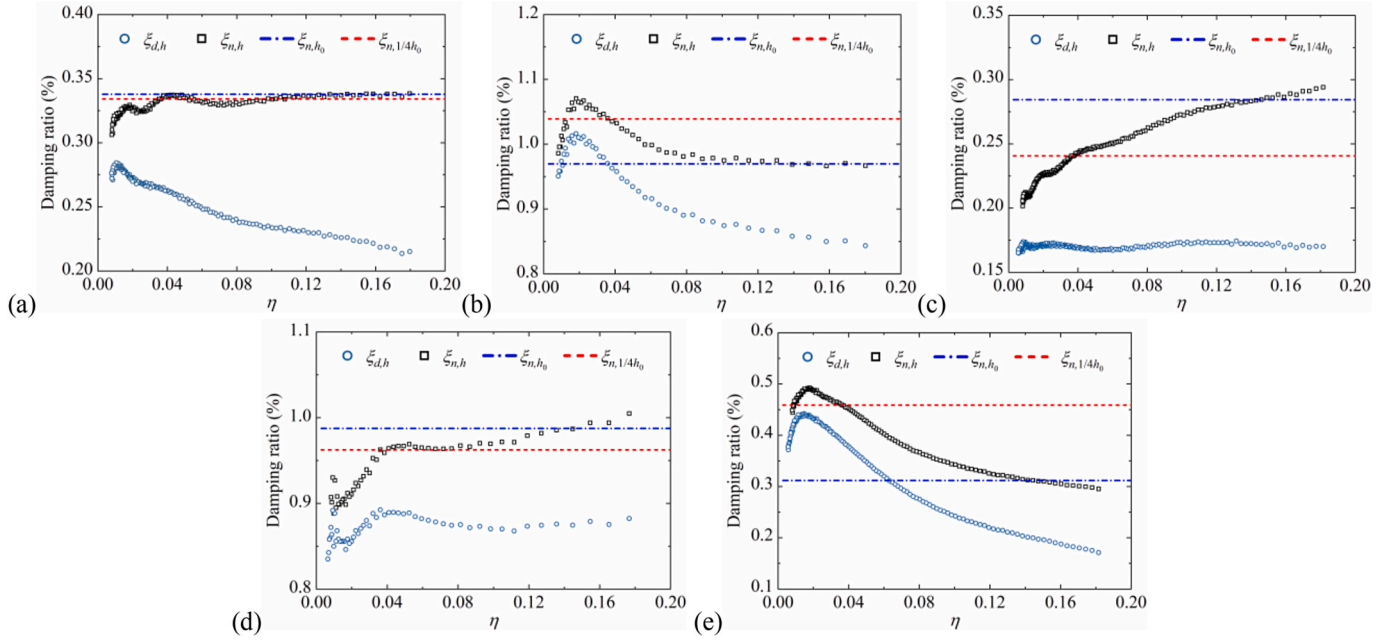


Fig. 17. Total mechanical damping ratios of the SSSM system under different damping adjustment methods: (a) damping condition 1; (b) damping condition 2; (c) damping condition 3; (d) damping condition 4; (e) damping condition 5.

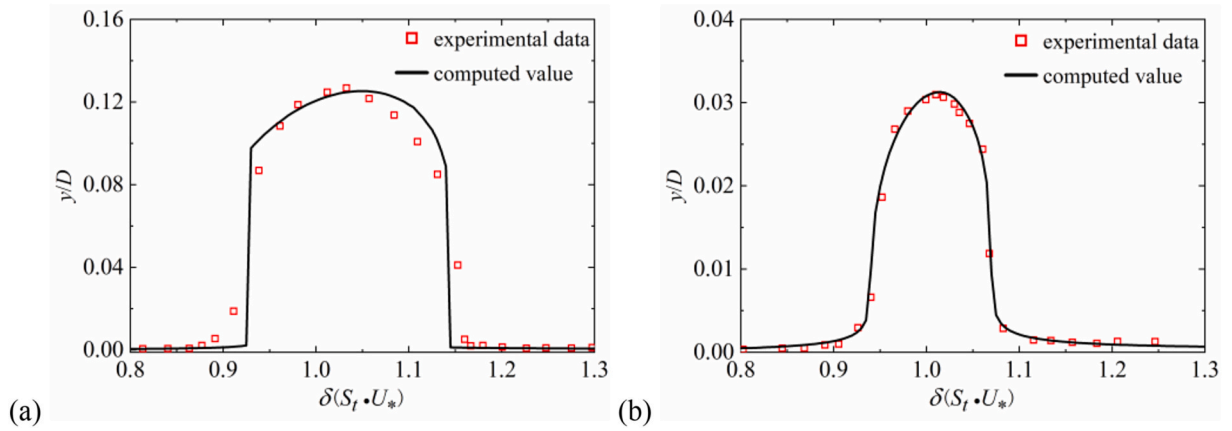


Fig. 18. Vertical VIV response of the Π -shaped composite girder under two levels of structural damping: (a) under the original mechanical damping; (b) under damping condition 4.

shown in Fig. 17, the effects of the nonlinearity of structural damping on the VIV responses of a streamlined box girder are further analyzed. It should be noted that the modal frequency of the streamlined box girder was not provided by Xu et al. (2015). Consequently, the value of h_0 cannot be directly calculated. Hence, the value of h_0 of the Π -shaped composite girder calculated in Section 4.3 is utilized for analysis. In Xu et al. (2015), velocity coupling between the wake vibrator and the main girder is adopted, i.e., $f(Y, Y', Y'') = FY'$. The key structural and aerodynamic parameters in Eq. (4) are as follows: $\xi = 0.005$, $St = 0.145$, $M = 0.0006315$, $\tilde{C}_{L0} = 0.2022$, $H_1^* = -1.6648$, $G = 0.5365$, and $F = 7.9908$.

By combining Eq. (4) with the above parameters, the vertical VIV responses of the streamlined box girder under each damping condition in Fig. 17 are calculated using the 4th-order Runge-Kutta method. The calculation results are shown in Fig. 20. As can be seen, except under damping conditions 1 and 4, there are obvious differences in the lock-in regions corresponding to the amplitude-dependent structural damping and the constant structural damping. The primary distinction lies in the cut-off wind speeds of these lock-in regions, and the cut-off wind speed in the lock-in region generally increases as the structural damping

decreases. For example, under the damping conditions 2 and 3, the amplitude-dependent structural damping ratios corresponding to the small VIV amplitudes of the main girder are smaller than the 'specification damping ratio' ξ_{n,h_0} and the 'small-amplitude damping ratio' $\xi_{n,1/4h_0}$, resulting in larger cut-off wind speeds. Under damping condition 5, the amplitude-dependent structural damping ratios corresponding to the small VIV amplitudes of the main girder fall between the 'specification damping ratio' ξ_{n,h_0} and the 'small-amplitude damping ratio' $\xi_{n,1/4h_0}$, leading to a cut-off wind speed that also falls between the two. The reason why the VIV responses of the main girder drop sharply at the cut-off wind speed or drop slowly near the cut-off wind speed is similar to the jumps and cliffs phenomena described in Section 4.3.

For the maximum VIV amplitudes of the main girder in Fig. 20, there are obvious differences between the results corresponding to the amplitude-dependent structural damping and the constant structural damping under damping conditions 2 to 5. The reason is that the maximum VIV amplitude of a structure generally increases with the decrease of the structural damping (Marra et al., 2015). For example, under damping conditions 2 and 4, the amplitude-dependent structural

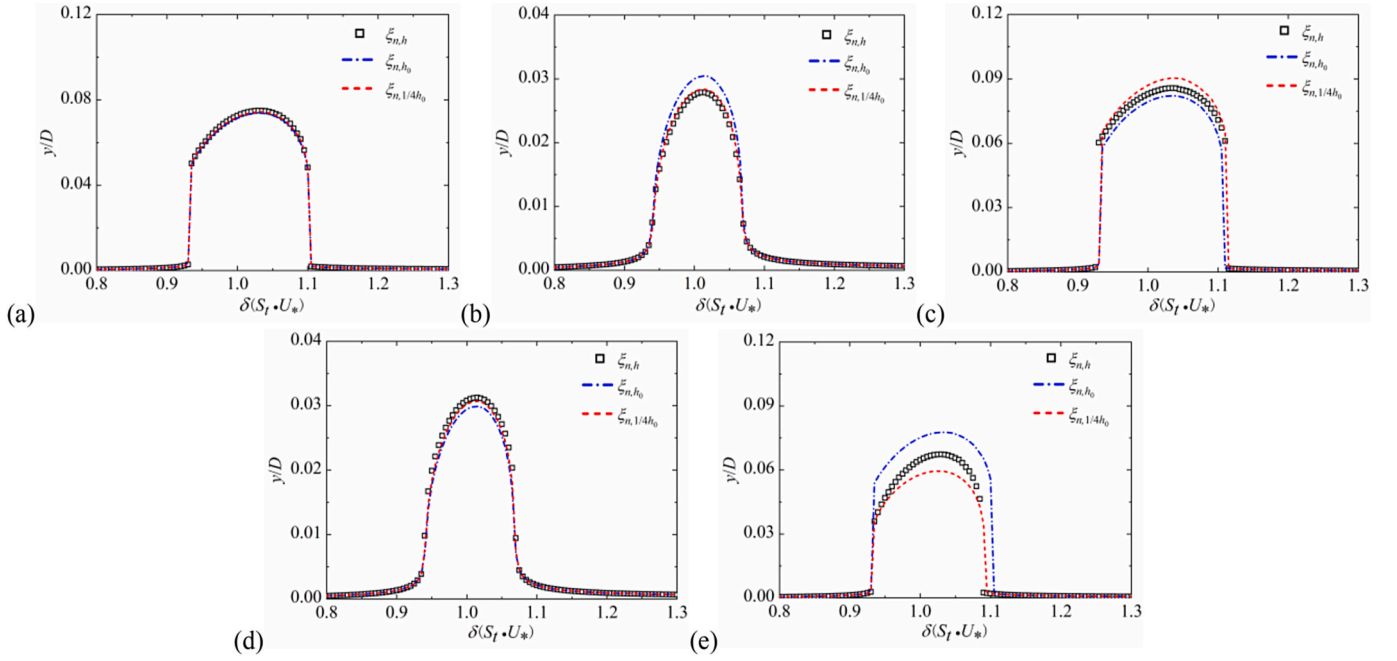


Fig. 19. Vertical VIV responses of the I-shaped composite girder under different damping adjustment methods: (a) damping condition 1; (b) damping condition 2; (c) damping condition 3; (d) damping condition 4; (e) damping condition 5.

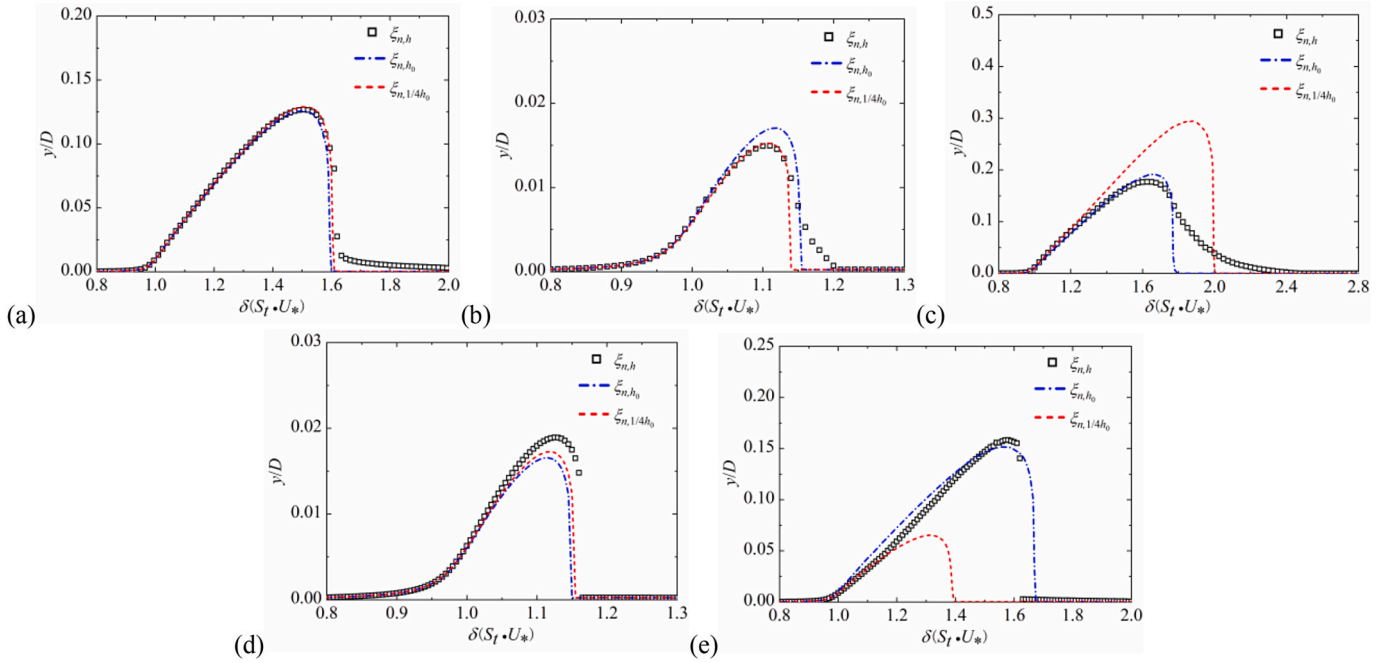


Fig. 20. Vertical VIV responses of the streamlined box girder under different damping adjustment methods: (a) damping condition 1; (b) damping condition 2; (c) damping condition 3; (d) damping condition 4; (e) damping condition 5.

damping ratios corresponding to the maximum VIV amplitudes of the main girder are close to the 'small-amplitude damping ratios' $\xi_{n,1/4h_0}$, so the corresponding maximum VIV amplitudes are also closer to the results corresponding to the 'small-amplitude damping ratios' $\xi_{n,1/4h_0}$. Under damping conditions 3 and 5, the amplitude-dependent structural damping ratios corresponding to the maximum VIV amplitudes of the main girder are close to the 'specification damping ratios' ξ_{n,h_0} , so the corresponding maximum VIV amplitudes are also closer to the result corresponding to the 'specification damping ratios' ξ_{n,h_0} . Therefore, when the nonlinearity of the structural damping is strong, the maximum

VIV amplitude of the main girder mainly depends on the value of the amplitude-dependent structural damping ratio at the maximum VIV amplitude.

5. Nominal value of the amplitude-dependent structural damping ratio

According to the research in Section 4, when a constant structural damping ratio is used to replace the amplitude-dependent structural damping ratio, there may be a significant difference in the maximum

VIV response of the main girder. To reduce this difference, the nominal value of the amplitude-dependent structural damping ratio is proposed in Section 5.1 and verified through wind tunnel tests in Section 5.2.

5.1. Method for defining the nominal value

The total mechanical damping ratios of the SSSM system corresponding to the three damping adjustment methods with a common intersection point are illustrated in Fig. 21. Among these, the total mechanical damping ratio provided by winding electrical tape matches the result in Fig. 17(a). The total mechanical damping ratios corresponding to the other two damping adjustment methods are consistent with those total mechanical damping ratios adjusted by increasing or decreasing a constant damping ratio in Fig. 17(c) and (e), respectively. The amplitude at the intersection point is 0.0751, which corresponds to the maximum VIV amplitude of the model when using the damping adjustment method of winding electrical tape. Based on the wake oscillator model identified in Section 4.3, the vertical VIV responses of the Π -shaped composite girder under the three different forms of total mechanical damping ratios in Fig. 21 are calculated using the 4th-order Runge-Kutta method. These calculation results are shown in Fig. 22. As can be seen, the variation curves for the three damping methods have the same peak point, i.e., their maximum VIV amplitudes are identical. To reduce the influence of damping nonlinearity on the maximum VIV amplitude of the main girder, this paper proposes to select the total mechanical damping ratio corresponding to the maximum VIV amplitude as the nominal value of the amplitude-dependent structural damping ratio in sectional model wind tunnel tests. Additionally, for a bridge deck, there may be several lock-in regions of VIVs or no VIVs at all. In the case of multiple lock-in regions of VIVs, it is recommended to take the total mechanical damping ratio at the maximum VIV amplitude of the main girder within the most unfavorable lock-in region as the nominal value. For the case where no VIVs occur, take the structural damping ratio at a certain small amplitude (for example, $\eta = 0.04$) as the nominal value.

In wind tunnel tests, the SSSM system can only simulate the dynamic characteristic parameters of the main girder in a specific low-order mode. Therefore, the variation of maximum VIV amplitude with structural damping ratio or Sc number based on sectional model tests is often used to predict the maximum VIV amplitudes of the main girder in other modes. In the above analysis, the total mechanical damping is obtained by excluding the windless-air-induced damping from the measured total damping of the SSSM system. To further investigate the effect of the windless-air-induced damping on the predicted maximum VIV amplitude, the VIV response of the Π -shaped composite girder is first calculated at the total mechanical damping. Then, based on the

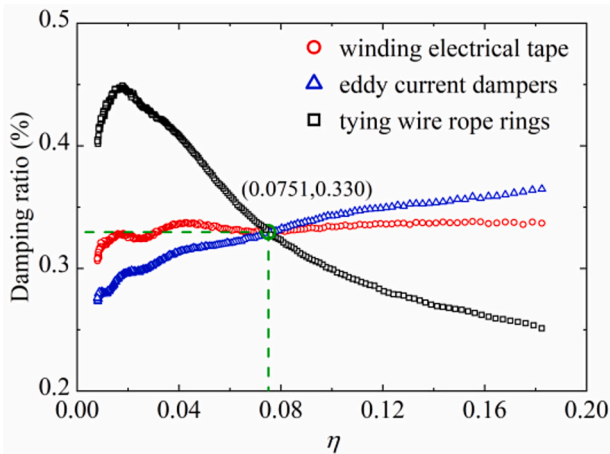


Fig. 21. Three forms of total mechanical damping ratios with a common intersection point.

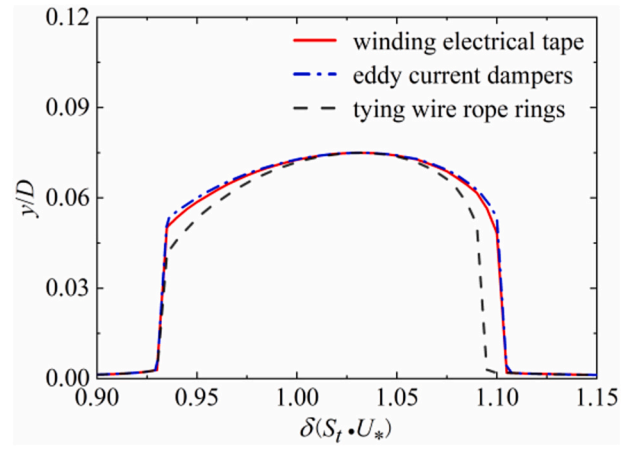


Fig. 22. Vertical VIV responses of the Π -shaped composite girder under three forms of total mechanical damping ratios.

computational results, the variation of the maximum VIV amplitude with the Sc number is obtained for two scenarios. The first scenario calculates the Sc number using the total mechanical damping ratio, while the second uses the total damping ratio, which combines the total mechanical damping ratio and the windless-air-induced damping ratio, as shown in Fig. 23. The horizontal coordinate is the nominal Sc number, $Sc = 4\pi m \xi_{nom} / (\rho B D)$, where ξ_{nom} is the nominal value of the structural damping ratio. As can be seen, when the nominal Sc number exceeds 8.62, the error in the influence of windless-air-induced damping on the prediction result remains within 10 %. Moreover, when the nominal Sc number surpasses 19.5, the error reduces to within 5 %. Therefore, when the structural damping is significant, the influence of the windless-air-induced damping on the predicted maximum VIV amplitude of the main girder can be disregarded. However, when the structural damping is low, it is imperative to exclude the windless-air-induced damping to ensure the accuracy of the prediction result.

5.2. Rationality verification of the definition method

To verify the rationality of the definition for the nominal value of the amplitude-dependent structural damping ratio, two damping adjustment methods are selected to adjust the total mechanical damping ratio of the large-scale SSSM system in Fig. 2(a), and wind tunnel tests are carried out. These two damping adjustment methods include installing eddy current dampers and tying wire rope rings, and the additional damping ratios provided by them are quite different in amplitude-dependent characteristics. The setup conditions for the sectional

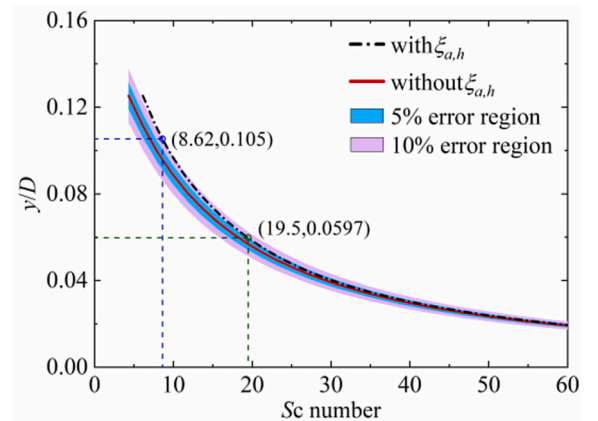


Fig. 23. Maximum VIV amplitude of the Π -shaped composite girder versus Sc number.

model test align with the parameters of the large-scale SSSM system, as presented in Table 1, Section 2.2. Details on the installation of eddy current dampers can be found in Section 3.2. Adjusting the air gaps allows for precise and continuous control of the total mechanical damping ratio of the SSSM system. The method of tying wire rope rings is identical to that described in Section 3.3, except that the diameter of the wire rope rings is 350 mm. The experimental procedure is as follows: firstly, adjust the total mechanical damping of the SSSM system using the damping adjustment method of tying wire rope rings and conduct a wind tunnel test to obtain the corresponding maximum VIV amplitude $A_{\max}$ of the Π -shaped composite girder. Next, readjust the total mechanical damping of the SSSM system using the damping adjustment method of installing eddy current dampers to equalize the total mechanical damping ratios of the SSSM system under both damping adjustment methods at the maximum VIV amplitude $A_{\max}$. Finally, conduct a wind tunnel test on the SSSM system under the damping adjustment method of installing eddy current dampers to obtain the corresponding VIV responses.

The vertical total mechanical damping ratios of the SSSM system under the two damping adjustment methods are illustrated in Fig. 24. As can be seen, there is an intersection point $P(0.0528, 0.541)$, i.e., when the dimensionless amplitude is 0.0528, the vertical total mechanical damping ratios for the two damping adjustment methods are the same, both are 0.541 %.

The vertical VIV responses of the Π -shaped composite girder under the two damping adjustment methods are illustrated in Fig. 25. As can be seen, the maximum VIV amplitudes under the two damping adjustment methods are basically the same, which are close to the amplitude of the intersection point P . The experimental results thus validate that employing the proposed nominal value definition method for structural damping ratios effectively mitigates the influence of the nonlinearity of the structural damping on the predicted maximum VIV amplitude of the main girder.

6. Conclusions

This paper presents an experimental and numerical study on the effects of the nonlinearity of structural damping on the VIV responses of bridge decks. The nonlinear damping characteristics of a SSSM system with/without different additional damping devices are obtained through a series of free decay vibration tests. The VIV responses of an open-section girder and a streamlined steel box girder under the linear and nonlinear structural damping are predicted and compared by employing a wake oscillator model, respectively. A nominal value for the amplitude-dependent structural damping ratio is also proposed. The main conclusions are summarized as follows:

- (1) The supplemental damping of the SSSM system by different devices is quite different: the eddy current dampers provide linear viscous damping; wire rope rings on the coil spring provide strongly nonlinear damping, which causes the total mechanical damping ratio of the SSSM system to gradually decrease with increasing amplitude; the electrical tape on the coil spring also provides nonlinear damping, whose magnitude increases with the number of tape turns and the width of the tape.
- (2) The amplitude and lock-in region of VIV for nonlinear structural damping are different from those for linear damping. These differences depend not only on the damping characteristics of the structure but also on the shape of the girder. For the specified streamlined box girder, its oscillation is highly sensitive to damping nonlinearity in terms of both the maximum VIV amplitude and the lock-in region. For the specified Π -shaped composite girder, the maximum VIV amplitude is sensitive to the damping nonlinearity, while the lock-in region is less affected.
- (3) Since the maximum amplitude is generally used to evaluate the performance of a bridge girder under VIV, it is suggested that the

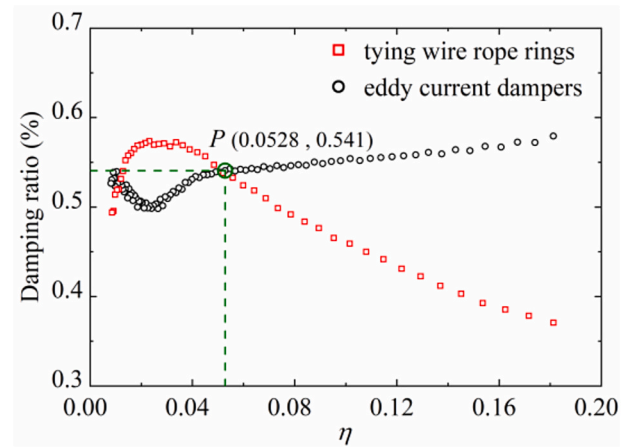


Fig. 24. Vertical total mechanical damping ratios of the SSSM system under two damping adjustment methods.

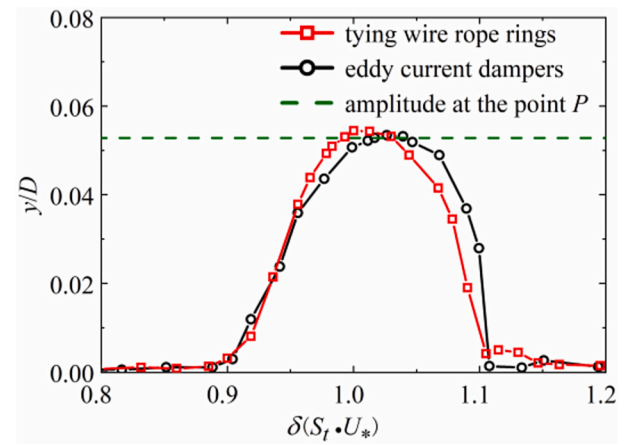


Fig. 25. Vertical VIV responses of the Π -shaped composite girder under two damping adjustment methods.

total structural damping ratio corresponding to the maximum VIV amplitude should be defined as the nominal value for the nonlinear structural damping. As a result, the energy dissipated by this nominal structural damping equals that input by the flow during the period of the largest VIV oscillation.

- (4) It is verified from sectional model wind tunnel tests that a SSSM system experiences the same maximum amplitude of VIV in different cases of nonlinear mechanical damping if they have the same nominal damping.

CRedit authorship contribution statement

Weimeng Ma: Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Data curation. **Zhiwen Huang:** Writing – review & editing, Visualization, Validation, Methodology, Formal analysis. **Xugang Hua:** Writing – review & editing, Validation, Methodology, Formal analysis, Conceptualization. **Zhengqing Chen:** Writing – review & editing, Validation, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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